The Invisible Universe

From dark matter to antimatter: a complete guide to modern physics and cosmology

Physics & Cosmology: A Complete Guide to Key Concepts

A comprehensive guide spanning the universe's composition, Einstein's relativity, the nature of light and photons, quantum mechanics, subatomic particles, black holes, antimatter, and nuclear energy. From the invisible scaffolding of dark matter to the violent annihilation of antimatter, this briefing connects the deepest ideas in modern physics into a single, coherent narrative.

DARK MATTER | RELATIVITY | QUANTUM MECHANICS | BLACK HOLES | ANTIMATTER | FUSION & FISSION

Part 1: The Universe's Composition

1.1 The Three Components of the Universe

The universe we inhabit is far stranger than it appears. Everything visible: every star, galaxy, planet, and atom we have ever detected, accounts for a mere sliver of what actually exists. According to NASA, the cosmos is divided into three fundamentally different components, each playing a distinct role in the structure and fate of the universe.

~5% Ordinary Matter

Everything we can see, touch, and interact with. Atoms, molecules, planets, stars, and all living things. The entirety of human knowledge about the physical world is built on this thin 5% slice.

~27% Dark Matter

An invisible, unknown substance that does not emit, absorb, or reflect light. Detectable only through its gravitational influence on ordinary matter and light. Its true nature remains one of the greatest unsolved mysteries in science.

~68% Dark Energy

A mysterious repulsive force embedded in the fabric of space itself, responsible for driving the universe's accelerating expansion. Even less understood than dark matter - we know it exists, but not what it is.

The extraordinary implication is that 95% of the universe is made of substances we cannot directly observe, measure, or explain with current physics. The visible cosmos, the subject of millennia of human astronomy, is merely the foam on the surface of a far deeper ocean.